

Hong Kong Metropolitan University School of Science and Technology

BSc. (Hons) in Data Science and Artificial Intelligence

**COMP S461F Data Science Project
2025-26**

Image-to-image Translation: Pixel-Art Generation with Transparency Background

by

**Ho Chun Long 13137320
Au Pak Yu 13136451
Chan Hoi Lun 12960272
Chan King Yin 13130121
Yuen Ching Hin 13133101**

**Supervisor: Mr. Kang XuDong
Date: 24 April 2026**

Abstract

Pixel art remains a prominent visual style across various domains, such as the gaming industry. However, creating high-quality assets remains a labour-intensive manual process, and there is a lack of automated generative tools specifically optimized for high-quality pixel art production-particularly when requiring precise transparency and pixel-level constraints. This project proposes an automated pixel art generator based on a modified pix2pix conditional GAN framework, designed to transform real-world photographs directly into pixel art with clean, transparent backgrounds.

To achieve high-fidelity outputs, we modified the U-Net generator architecture to handle low-resolution synthesis (32×32) while introducing a specialized transparency control mechanism. By decoupling color and alpha channels and applying a custom Binary Regularization loss, the model ensures sharp, "all-or-nothing" transparency (Alpha 0 or 1), eliminating the common issue of semi-transparent artifacts.

Experimental results demonstrate the model's effectiveness. In a human preference study, our approach was favoured in 80% of cases compared to baseline CNN and standard pix2pix models. Furthermore, quantitative metrics show a superior LPIPS score (0.0484) with high training stability. This work provides a practical tool for indie developers and rapid prototyping by simultaneously addressing automated generation and background removal.

Table of Contents

Chapter 1 – Introduction	5
1.1 Background	5
1.2 Field/Concepts	5
1.3 Problem Statement	5
1.4 Objectives	6
Chapter 2 – Literature Review	6
2.1 Development of GANs	6
2.2 Overview of Alternative Deep Generative Models	9
Chapter 3 – Method Design	9
3.1 Dataset	9
3.2 Model Design	10
3.2.1 U-Net Architecture	10
3.2.2 Output Layer Design for RGBA	11
3.2.3 Training Strategy	12
3.3 Evaluation Method	14
3.3.1 CNN vs Our model	14
3.3.2 Original pix2pix vs Our model	15
3.4 Experimental Design	15
3.4.1 Qualitative Evaluation	15
3.4.2 Quantitative Evaluation	15
Chapter 4 – Results & Analysis	17
4.1 Qualitative Results	17
4.1.1 Original pix2pix	17
4.1.2 Baseline CNN	18
4.1.3 Our model	18
4.1.4 Qualitative Evaluation Result	19
4.2 Quantitative Evaluation Result	19
4.2.1 Fréchet Inception Distance (FID Score)	19
4.2.2 Learned Perceptual Image Patch Similarity (LPIPS)	20
Chapter 5 – Discussion	20
5.1 Observations Discussion	20
5.1.1 Why Does Our Model Achieve Better Performance?	20
5.1.2 Limitation of FID in Pixel-Style Image Generation	21
5.1.3 Why is LPIPS a More Suitable Metric for Our Task?	21
5.1.4 Suitability of the PatchGAN discriminator	21
5.2 Reflection	22

5.2.1 The Multi-View Challenge.....	22
5.2.2 Limitations and Scope of Multi-View Generation.....	22
5.2.3 Key Insights: Bridging Theory and Practice	22
5.3 Future Work.....	23
5.3.1 Implement Learned Perceptual Loss for Improving Pixel-Art Sharpness	23
5.3.2 Multi-view Generation	23
Chapter 6 – Conclusion	23
Work Division	25
Reference	26

Chapter 1 – Introduction

1.1 Background

In recent years, pixel-style games have surged in popularity. However, the game industry is still facing a significant challenge: the lack of efficient, automated tools for generating high-quality pixel art assets (Shinkle, 2021). As demand for pixel-style games continues to surge, the traditional manual workflow has become a major bottleneck.

Creating pixel art assets remains highly labour-intensive and costly. Artists must manually remove backgrounds and strictly enforce limited colour palettes. This repetitive and time-consuming process significantly increases both development time and budget, making it particularly difficult for small indie teams and rapid prototyping. Historically, even large studios such as Sega in the 1980s and 1990s relied on custom hardware and painstaking hand-drawing techniques to produce pixel art, highlighting how resource-intensive the craft has always been (Isidore, 2016).

According to a gaming report conducted by Bain & Company in 2025 (Plantec & Christofferson, 2025), AAA game studios are increasingly caught between highly agile indie developers and large platform-style games. They face rising development costs, shrinking margins, and intensifying competition. In this environment, the ability to quickly and cost-effectively generate consistent pixel assets has become a critical competitive advantage.

Considering this, our research explores the application of deep learning as a solution to automate the pixel art creation process, thereby reducing the manual workload and financial cost for developers.

1.2 Field/Concepts

Deep learning is a well-known domain of machine learning and artificial intelligence that uses multi-layer neural networks to compute and analyse complicated patterns in large datasets, especially in high-dimensional data tasks, like images, audio, and natural language.(McKinsey, 2024) Its ability to learn hierarchical representations and integrate multiple processing steps into a single end-to-end pipeline makes it particularly effective for addressing labour-intensive tasks in game asset creation.

In computer vision, image generation is a key task that focuses on synthesizing new, visually convincing images rather than merely analysing existing ones. This capability underpins applications such as style transfer and image-to-image translation, which are highly relevant to creative asset production. Generative Adversarial Networks (GANs), a powerful deep learning-based generative framework, were developed to tackle this challenge (Gulrajani et al.,2017). In a GAN, a generator learns to produce realistic images while a discriminator learns to distinguish real images from generated ones. The two networks train adversarially, pushing the generator to create increasingly convincing outputs.

1.3 Problem Statement

Although pixel art remains a widely used visual style in gaming, producing high-quality pixel art assets is still a highly labour-intensive manual process. Clean transparency (a precise alpha channel) is one of the critical requirements for these assets, so it enables seamless integration into game engines and avoid visual artifacts. Game engines such as Unity or Unreal Engine

rely on correct alpha channels to render sprites correctly. A clean alpha ensures the engine to draw only the intended pixels without any unwanted background data.

Despite this need, most mainstream tools fails to satisfy both production requirements at once: (1) pixel art style including limited color palettes, accurate dithering, crisp edges, and consistent resolution - and (2) transparent background (clean alpha ready for engines). Where partial solutions exist, they often require stitching together multiple tools or relying on overly complex, heavyweight models that are slow to process and difficult to integrate into real production workflows.

Existing tools and methods fall short in several ways. General-purpose image generation models often produce outputs with incorrect resolution, broken pixel-level details, or poor semantic fidelity when converting real-world references into pixel art. Even specialized approaches typically address only one aspect (e.g., style transfer or background removal) but not both, forcing artists to combine multiple tools in complex, time-consuming pipelines. Moreover, many solutions rely on large, heavyweight models that are computationally expensive, difficult to fine-tune, and impractical for iterative production workflows.

1.4 Objectives

To sum up, the objectives of this project are summarized as below:

- **Generate High-Fidelity Pixel Art Style:** Develop a model capable of converting real photographs into high-quality pixel art GAN techniques, focusing on accurate colour palette management, efficient dithering, and sharp pixel edges.
- **Produce Fully Transparent Background:** Convert input images with backgrounds into pixel art outputs with fully transparent background, successfully removing the original background while preserving the main subject with clean and sharp edges.
- **Develop and implement a system:** Combines a learnable alpha matting head with a conditional GAN architecture to forecast backdrop transparency and pixel art stylization in a single forward pass, avoiding multi-stage pipelines.

Chapter 2 – Literature Review

2.1 Development of GANs

2014 — GAN: The Starting Point

Goodfellow et al. (2014) introduced Generative Adversarial Networks (GANs), where a generator competes against a discriminator in a minimax game to synthesize realistic samples without explicit likelihood modelling.

Early GANs proved that adversarial learning can capture rich image distributions, but they suffered from training instability (mode collapse, vanishing gradients) and, crucially for production use, they were unconditional—offering little control over what gets generated. However, our task requires conditional image-to-image translation, where a real photograph is transformed into a corresponding pixel art image. The original GAN is not available for accepting an input image as condition, making it unsuitable for our paired translation objective.

2014 — Conditional GAN (cGAN): Adding Control

$$L_{cGAN}(G, D) = E_{x, y} [\log D(x, y)] + E_{x, z} [\log (1 - D(x, G(x)))]$$

Where:

- G : The Generator, which attempts to learn the mapping from input x and noise z to the target output y
- D : The Discriminator, which learns to distinguish between real pairs x, y and synthesized pairs $(x, G(x, z))$

Mirza and Osindero et al. (2014). extended GANs by conditioning both generator and discriminator on side information y (e.g., class labels, images, or text). Based on GAN, cGAN added conditioning information y . This conditioning mechanism provides explicit control over the generation process. It allows the model to learn a mapping from the noise and the conditional to the target output distribution. For example, beyond basic labels, conditions can be based on geometric attributes, e.g. rotation angles, so that the model produces variations of an object at some specified point of view. It is possible to include encoded angle values into the conditioning vector y , thus enabling the generator to produce rotated images without a three-dimensional model. This makes the generation controllable and allows for more variables in translating images. The conditional mechanism allows the generator to take a real image as input and produce a corresponding stylized output, which satisfies our needs. However, the original cGAN still suffers from mode collapse and training instability.

2015 — Deep Convolutional GAN (DCGAN): A Stable Backbone for Images

Radford et al. (2015) proposed DCGAN, replacing Multilayer Perceptron (MLPs) with convolution and transpose-convolution layers, which are more suitable for image processing. The architecture also introduced Batch Normalization and appropriate activation functions. DCGAN became the practical backbone for image GANs and many conditional variants. Although DCGAN exhibits improved training stability relative to the original generative adversarial network, it remains susceptible to vanishing gradient and mode collapse phenomena. Although DCGAN exhibits improved training stability, it remains susceptible to vanishing gradient and mode collapse phenomena. These issues suggest that the original DCGAN architecture, despite its convolutional backbone, lacks sufficient representational power for highly complex or multi-modal image distributions.". More critically, DCGAN is an unconditional architecture which is not designed for paired image-to-image translation tasks. It synthesises images solely from random noise vectors and lacks the capacity to accept an input image as a conditioning variable to produce a corresponding output.

2016–2018 — Creating Scalable Trainable GANs

To address instability and collapse in adversarial training, several techniques were proposed and widely adopted:

- WGAN: Arjovsky, Chintala, and Bottou (2017) reframed the objective with the Wasserstein/Earth-Mover distance to make the gradients smoother and less mode collapse.
- WGAN-GP: Gulrajani et al. (2017) replaced crude weight clipping with a gradient

penalty, enforcing the smoothness of the gradient descent and improving training stability.

- LSGAN: Mao et al. (2017) substituted the adversarial cross-entropy with a least-squares loss, alleviating generator vanishing gradients
- Spectral Normalization Miyato et al. (2018) applied spectral norm regularization to discriminator weights to control the Lipschitz constant, further stabilizing learning and improving sample quality. regularizes discriminator weights and further stabilizes learning.

2016/2017 — pix2pix: cGAN + DCGAN for Image-to-Image Translation

$$G^* = \arg \min_G \max_D [\mathcal{L}_{cGAN}(G, D) + \lambda \mathcal{L}_{L1}(G)]$$

Where $\mathcal{L}_{cGAN}(G, D)$ refers to the cGAN objective function

Isola et al. (2016) reframed the foundations of cGAN for conditional generation as paired image-to-image translation. It's architectural backbone relies heavily on the principles of DCGAN.

Relationship with cGAN: Unlike a traditional GAN that maps a random noise vector z to an output, pix2pix operates as a cGAN. The input image x serves as a structural prior. Both the Generator G and the Discriminator D are "conditioned" on this input, ensuring that the generated output is not just a realistic image, but one that strictly adheres to the spatial layout of the source.

Relationship with DCGAN: Pix2pix adopts the all-convolutional design of DCGAN, replacing deterministic pooling layers with strided convolutions (discriminator) in the encoder and fractional-strided convolutions (generator) in the decoder. This allows the model to learn its own spatial downsampling and upsampling, which is crucial for preserving the structure of images during translation.

Given an input image (condition) and its paired target, pix2pix also introduces a few more tricks to improve the quality and fidelity of the generated results:

- PatchGAN discriminator: judge realism at the patch level which is excellent for textures, edges, and local structure.
- Skip Connection (concatenation): shuttle low-level features from encoder to decoder to preserve fine-grained spatial details.
- Composite loss (adversarial + L1 reconstruction loss): ensure realism and structural faithfulness.

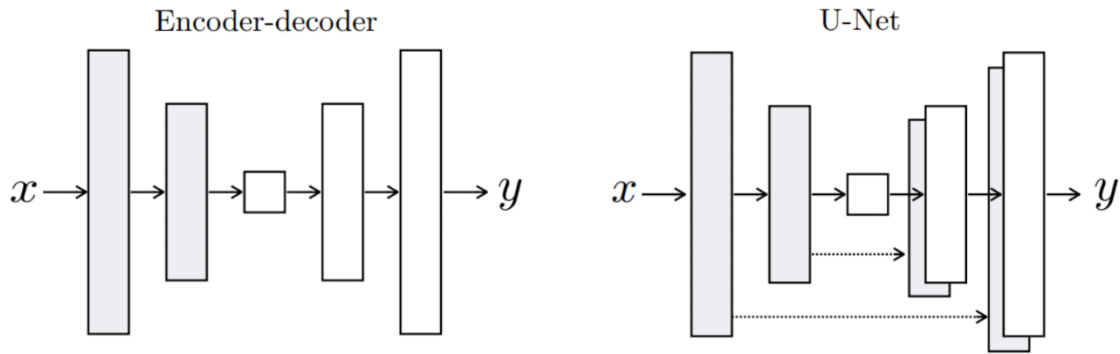


Figure 1: Architecture of pix2pix (Image-to-Image Translation with Conditional Adversarial Networks). Note. Adapted from Isola et al. (2017). Reprinted from <https://arxiv.org/pdf/1611.07004>

2.2 Overview of Alternative Deep Generative Models

Denoising Diffusion Probabilistic Models

Although GANs provide an efficient and lightweight framework well-suited for our paired image-to-image translation task, other deep generative approaches have also gained significant attention.

Ho, Jain, and Abbeel (2020) and Nichol and Dhariwal (2021) introduced Denoising Diffusion Probabilistic Models (DDPMs), which have since evolved into powerful diffusion-based frameworks. Models such as **Stable Diffusion (Rombach et al., 2022)** (and its successors) have become the dominant paradigm in general-purpose generative AI for image synthesis by 2025–2026. Unlike GANs, which generate images in a single forward pass through adversarial training, diffusion models work by iteratively denoising random noise guided by conditioning signals (e.g., text prompts or input images). This iterative refinement process typically yields higher perceptual quality, greater diversity, and more stable training, avoiding common GAN issues such as mode collapse.

While diffusion models are powerful for generative image, it is computationally heavier and slower inference compared to GAN model. Their strengths lie in high-resolution, open-ended, or photorealistic generation, which makes them somewhat overkill for our task that demands fast, deterministic, and highly controllable paired translation. although diffusion models such as Stable Diffusion represent the current mainstream in general generative AI, they are not the ideal choice for this project. Instead, GAN model will be considered.

Chapter 3 – Method Design

3.1 Dataset

Building on the literature review’s conclusion that the modified pix2pix CGAN provides the most efficient and suitable framework for this paired image-to-image translation task. We have found a paired dataset that can support our model training. This dataset directly aligns with the

primary purpose of the project: to develop a model capable of converting real-world animal photographs into high-fidelity pixel art while simultaneously removing the background to produce clean, fully transparent RGBA outputs. In doing so, the paired dataset enables the model to address the labour-intensive challenges in game asset creation.

The dataset used in this project consists of 1,171 paired animal images. Real images were collected from the Kaggle "Animals with Attributes 2" dataset (Kumar, 2023). Corresponding pixel art style images were processed using online pixel art tools. Each real image has its corresponding pixel art version, forming a paired dataset suitable for image-to-image translation tasks. This paired dataset enables the conditional GAN to simultaneously learn both pixel art stylization (accurate colour palette, efficient dithering, and sharp edges) and fully transparent background removal. The corresponding pixel art ground truths were carefully prepared with limited colour palettes and crisp edges to ensure high-fidelity results.

The dataset was randomly split into training, validation, and test sets with a ratio of 7:1.5:1.5 (70% training, 15% validation + 15% test). All images were resized to 128×128 resolution to match the input size of our modified U-Net model.

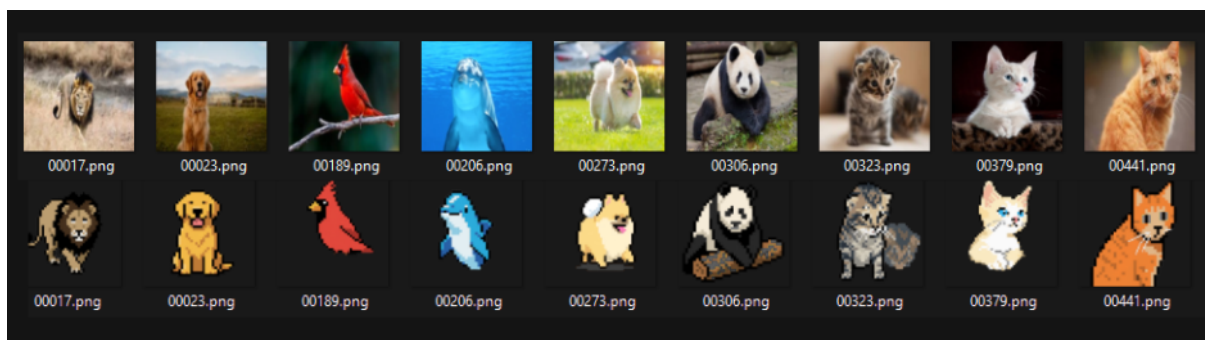


Figure 2: examples of paired data in our dataset at 1024×1024 resolution.

3.2 Model Design

3.2.1 U-Net Architecture

The original U-Net, proposed by Ronneberger et al. (2015) and widely used in pix2pix, consists of a symmetric encoder-decoder structure. It performs continuous downsampling from the input size (e.g. 128×128) all the way to a very small bottleneck (usually 1×1 or 2×2), followed by continuous upsampling back to the original resolution. Skip connections (concatenation) are applied between corresponding encoder and decoder layers to preserve the image spatial details and information. However, in our version, we modified the U-Net architecture to better suit the pixel art generation task (figure2). The model still starts with 128×128 RGB input images, but we only downsample to an 8×8 bottleneck instead of going all the way down to 1×1. After reaching the bottleneck, the decoder upsamples only up to 32×32 resolution with 4 RGBA channels ("A" represent Alpha channel which represents pixel opacity). Skip connections via concatenation are retained to help preserving important structural information. LeakyReLU is applied for the U-Net. Unlike the standard ReLU activation function, which suffers from the "Dying ReLU" problem — where neurons output zero for all negative inputs and become permanently inactive, significantly reducing the model's learning capacity. LeakyReLU introduces a small negative slope for negative input values. This allows a small and non-zero

gradient to flow through the network even for negative activations, thereby preventing dead neurons, and improving training stability. The mathematical definition of LeakyReLU is:

$$f(x) = \begin{cases} x & \text{if } x \geq 0 \\ \alpha x & \text{if } x < 0 \end{cases}$$

Where α is the coefficient for controlling the slopes for negative input values

Our design with low-resolution output (32x32) aligns well with the project target of generating crisp pixel art style images.

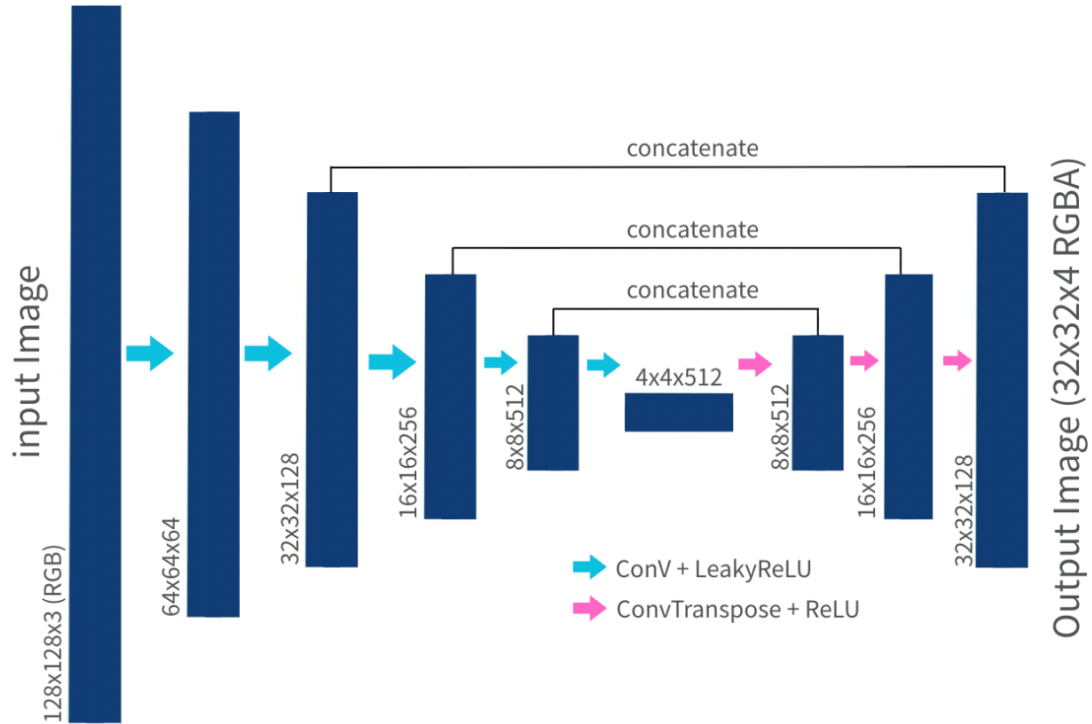


Figure3: Our designed U-net architecture (example for 8x8 pixels in the lowest resolution). Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on bottom of the box.

3.2.2 Output Layer Design for RGBA

To achieve a fully transparent background in the generated pixel art, the Alpha channel must output exactly 0 (full transparency). Therefore, we separated the final output layer into two independent branches:

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}, \quad -1 < \tanh(x) < 1$$

- **RGB channels:** Use hyperbolic tangent activation function (tanh function), which outputs values in the range [-1, 1]. This design matches the data processing pipeline: Original images are loaded with pixel values in [0, 1]. After normalization (mean = 0.5, std = 0.5), the input tensor is scaled to [-1, 1], which perfectly aligns with the Tanh activation for the RGB channels.

$$\sigma(x) = \frac{1}{1 + e^{-x}}, \quad 0 < \sigma(x) < 1$$

- Alpha channel: Use Sigmoid activation function, which outputs values in the range [0, 1]. Sigmoid function naturally produces outputs between 0 and 1, which is exactly the required range for the Alpha channel (where 0 means fully transparent and 1 means fully opaque).

The implementation logic is as follows:

1. Input: u3 (feature from last decoder layer)
2. RGB = tanh(conv1x1_rgb(u3))
3. Alpha = sigmoid(conv1x1_alpha(u3) x 5)
4. Output = concatenate(RGB, Alpha)

Different activation functions are deliberately chosen for RGB (Tanh) and Alpha (Sigmoid) to match their respective value ranges and solve the transparent background problem. In Alpha channel, the multiplication by 5 before sigmoid makes the function steeper, forcing the model to output values closer to 0 or 1 instead of staying around 0.5 (semi-transparent).

3.2.3 Training Strategy

Generator Loss

To generate high-quality pixel-art style image, we implement multiple losses to train our model.

Adversarial Loss

$$\mathcal{L}_{adv} = \text{BCE}(D(x, G(x)), 1)$$

Where:

- BCE is Binary Cross-Entropy with Logits Loss
- x is the real input image
- $G(x)$ is the generated image

Adversarial Loss encourages the generator to maximize the probability of the discriminator classifies the generated image as real.

L1 Loss for RGB

we designed a specialized L1 loss function that decouples the supervision of RGB colour and Alpha transparency channels. The RGB loss is formulated as:

$$\mathcal{L}_{L1} = \sum_{i=1}^N \| G(x)_{RGB} - y_{RGB} \|_1$$

Where:

- $G(x)_{RGB}$ is the RGB channel of generated image
- y_{RGB} is the alpha channel of ground truth

L1 Supervision for Alpha

To improve background removal, we introduce an explicit L1 supervision on the alpha channel between the generated output and the ground-truth alpha channel.

$$\mathcal{L}_{alpha_L1} = \sum_{i=1}^N \| G(x)_{alpha} - y_{alpha} \|_1$$

Where:

- $G(x)_{alpha}$ is the alpha channel of generated image
- y_{alpha} is the alpha channel of ground truth

Binary Regularization

During training, the sigmoid activation in the Alpha channel always predicts around 0.5 as the output value. This resulted in semi-transparent (grey) backgrounds instead of fully transparent background (Alpha = 0).

Solution:

To solve this problem, we introduced Binary Regularization into the Generator loss. This custom loss term encourages the Alpha channel to produce values close to 0 (fully transparent) or 1 (fully opaque).

$$\mathcal{L}_{alpha_bin} = \frac{1}{N} \sum_{i=1}^N \left(\alpha_{fake}^{(i)} \times (1 - \alpha_{fake}^{(i)}) \right)^2$$

Where:

- $\mathcal{L}_{alpha_bin}$ becomes smaller (almost 0) when alpha_fake is close to 0 or 1
- $\mathcal{L}_{alpha_bin}$ becomes larger, applying a strong penalty when alpha_fake stays around 0.5
- The mean operation averages the penalty across the entire batch.

This regularization effectively forces the model to make clearer decisions on transparency, successfully solving the issue of unwanted semi-transparent backgrounds in the generated pixel art.

Generator Total Loss

The generator is trained to fool the discriminator while simultaneously minimizing the distance to the ground truth. The generator loss is a weighted sum of four specific components. Based on empirical tuning, the final implemented total loss function is defined as:

$$\mathcal{L}_G = \mathcal{L}_{adv} + \lambda_{L1} \cdot \mathcal{L}_{L1} + \lambda_{alpha_L1} \cdot \mathcal{L}_{alpha_L1} + \lambda_{alpha_bin} \cdot \mathcal{L}_{alpha_bin}$$

Where:

- λ_{L1} represent the weight of $\mathcal{L}_{L1}$, which $\lambda_{L1} = 100$
- λ_{alpha_L1} represent the weight of $\mathcal{L}_{alpha_L1}$, which $\lambda_{alpha_L1} = 20$
- λ_{alpha_bin} represent the weight of $\mathcal{L}_{alpha_bin}$, which $\lambda_{alpha_bin} = 10$

Discriminator Loss

Discriminator is trained using the Binary Cross Entropy (BCE) loss with logits. Its objective is to correctly distinguish between real and fake image pairs. The loss function is defined as:

$$\mathcal{L}_D = \frac{1}{2} [\mathcal{L}_{BCE}(D(x, y), 1) + \mathcal{L}_{BCE}(D(x, \hat{y}), 0)]$$

Where:

- x represents the real input image (condition)
- y the ground truth pixel art image (with alpha channel)
- $\hat{y} = G(x)$ is the generated fake pixel art image produced by generator
- $D(x, y)$ is discriminator's output when given a real pair
- $D(x, \hat{y})$ is discriminator's output when given a fake pair

The BCE loss encourages the discriminator to output values close to 1 for real pairs and close to 0 for fake pairs. By using this adversarial formulation, the discriminator learns to critically evaluate both the pixel art style quality. This provides strong adversarial feedback to the generator, pushing it to produce more realistic outputs.

3.3 Evaluation Method

We compared our modified U-Net with a baseline CNN model and original pix2pix model to analyze the impact of key design choices.

3.3.1 CNN vs Our model

The details of baseline CNN models:

- Only Generator (no PatchGAN Discriminator)
- No skip connections: no concatenation between encoder and decoder layers
- ReLU activation function instead of LeakyReLU in both downsampling and upsampling process
- Same overall depth and channel progression for fair comparison (32×32 resolution with 4 RGBA channels)

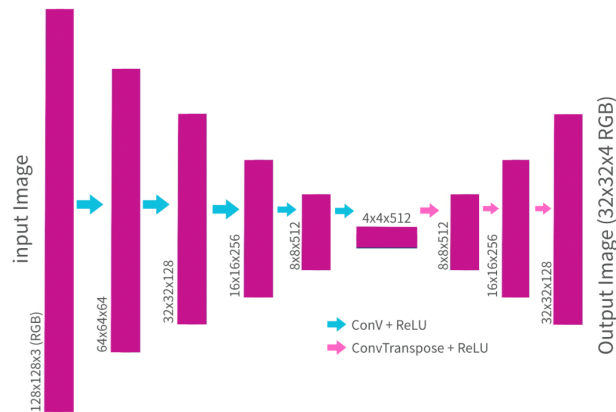


Figure 3: Baseline CNN model architecture (example for 8x8 pixels in the lowest resolution). Each pink box corresponds to a multi-channel feature map. The number of channels is denoted on bottom of the box.

3.3.2 Original pix2pix vs Our model

The details of original Pix2pix models:

- U-Net generator with PatchGAN Discriminator
- skip connections: concatenation between encoder and decoder layers
- LeakyReLU activation in both downsampling and upsampling process
- Upsampling to 128x128 instead of 32x32 in upsampling process

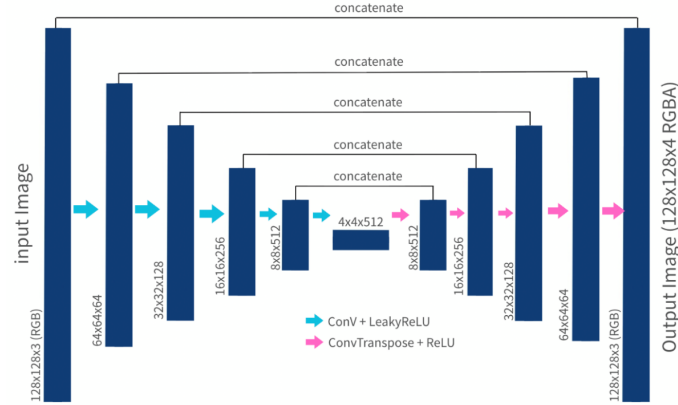


Figure 4: Original Pix2pix model architecture (upsampling to 128x128). Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on bottom of the box.

3.4 Experimental Design

3.4.1 Qualitative Evaluation

To evaluate the visual quality of the generated pixel art, we conducted a human preference study involving **20 independent candidates**. Each candidate was shown 100 randomly selected output images from three different models: our modified U-Net, the baseline CNN model, and the original Pix2pix model. The candidates were asked to choose which model's output they preferred most for each image based on overall visual quality, sharpness of edges, colour consistency, and pixel art style fidelity.

3.4.2 Quantitative Evaluation

1) Fréchet Inception Distance (FID Score)

Fréchet Inception Distance (FID), proposed by Heusel et al. (2017), is one of the most widely adopted metrics for evaluating the performance of generative models such as GANs. Instead of comparing images at the pixel level, FID measures the similarity between the statistical distributions of real and generated images in a high-dimensional latent feature space. It uses a pre-trained Inception-v3 network to extract 2048-dimensional feature vectors from the pool3 layer.

The mathematical formula:

$$FID\ Score = ||m - m_w||^2 + Tr(C + C_w - 2\sqrt{CC_w})$$

Where:

- $m - m_w$: Mean difference of target paired images features

- $C + C_w - 2\sqrt{CC_w}$: Covariance of target paired images features
- $\text{Tr}(\cdot)$: Trace of the matrix

Interpretation of FID Score:

- Lower FID = Better performance The generated images are statistically closer to real images in mean and covariance.
- Higher FID = Poorer performance Indicates noticeable differences in quality, diversity, texture, or structure

2) Learned Perceptual Image Patch Similarity (LPIPS)

Learned Perceptual Image Patch Similarity (LPIPS), proposed by Zhang et al. (2018), is a relatively new and widely recognized evaluation metric for assessing the perceptual quality of generated images. Unlike traditional pixel-wise metrics such as L1 or L2 loss, LPIPS was specifically designed to better align with human visual perception. It utilizes deep features extracted from well-known and well-pre-trained convolutional neural networks, primarily AlexNet and VGG, which have been trained on large-scale image classification tasks (e.g., ImageNet).

Instead of comparing images at the raw pixel level, LPIPS computes the distance between corresponding feature patches at multiple layers of these pre-trained networks. The core idea is that features learned by these deep networks can effectively simulate how humans perceive and judge image similarity. By measuring the difference in these perceptual features, LPIPS provides a more accurate reflection of visual quality as perceived by human observers.

LPIPS is particularly suitable for evaluating pixel art generation because it focuses on perceptual similarity instead of pixel-level accuracy, which aligns well with human judgment of pixel art quality.

The mathematical formula:

$$\text{LPIPS}(x, y) = \sum_l w_l \cdot \frac{1}{H_l W_l} \sum_{h,w} |\hat{f}_l(x)_{h,w} - \hat{f}_l(y)_{h,w}|_2^2$$

Where:

- $w_l = 1 \forall l$ is equivalent to computing cosine distance

Interpretation of LPIPS Score

- Lower LPIPS = Better performance The generated images are perceptually more similar to real images, aligning closely with human visual perception.
- Higher LPIPS = Poorer performance Indicates visible differences in sharpness, texture, edge quality, and overall visual naturalness.

Chapter 4 – Results & Analysis

4.1 Qualitative Results

4.1.1 Original pix2pix

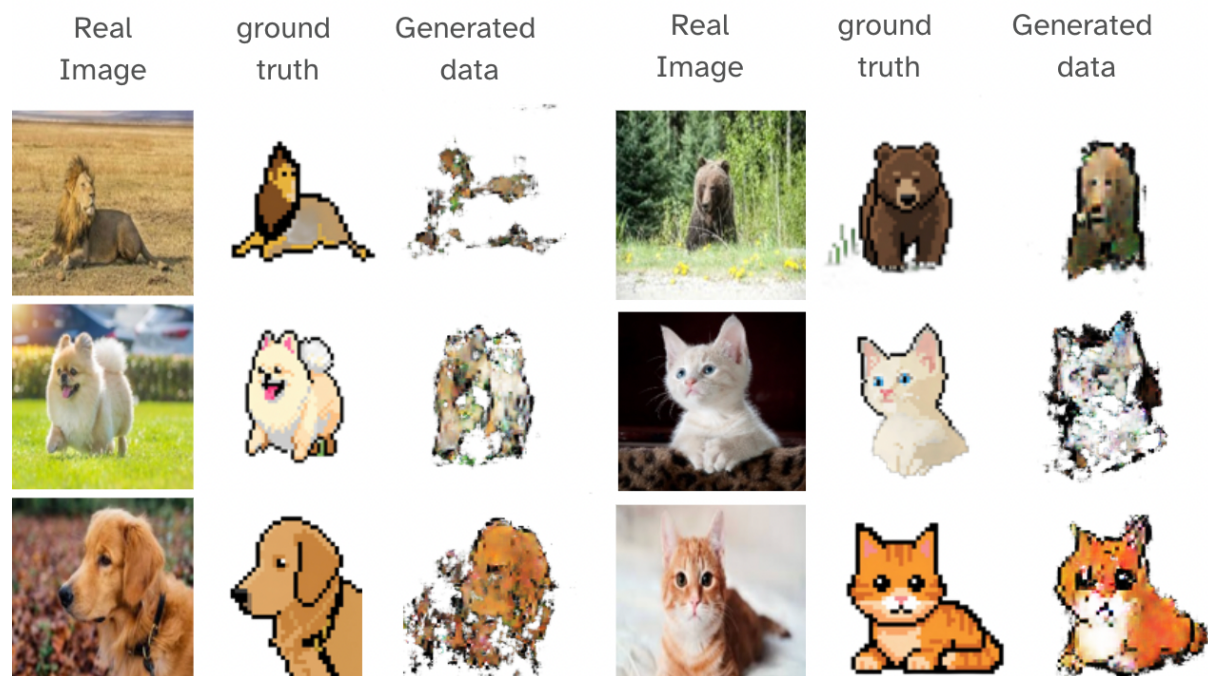


Figure 5: Output of Original pix2pix model

Figure 5 shown some examples of the results of the original model. While the model generates image from real images to pixel-art style by upsampling to 128x128 instead of 32x32, the generated images were far from ideal. The outputs showed blob-like appearances, fragmented boundaries, melted edges and local regions collapse into colour patches. From this, it is evident that the original model struggled on preserving details and maintaining clear object boundaries during the generation process.

4.1.2 Baseline CNN

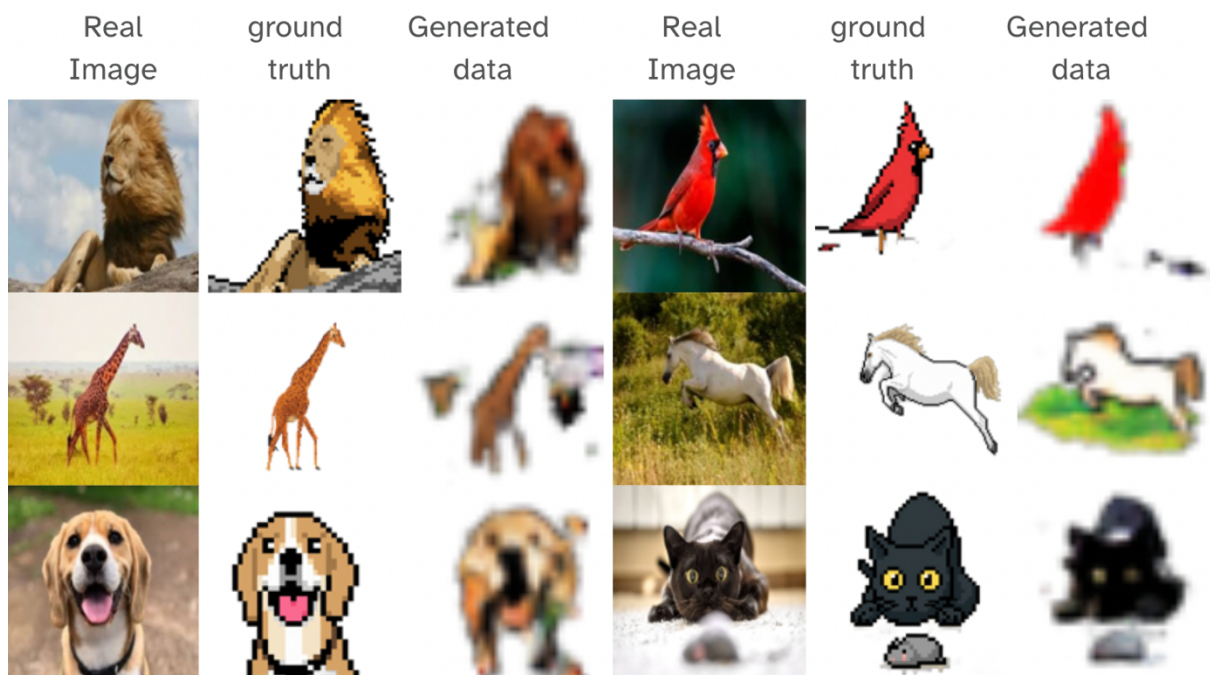


Figure 6: Baseline CNN model

In addition, Figure 6 shows the outputs of the baseline CNN model. The generated images are of very low quality, appearing mostly as blurred and fragmented blobs. The main features of the animals are heavily distorted or completely lost.

4.1.3 Our model

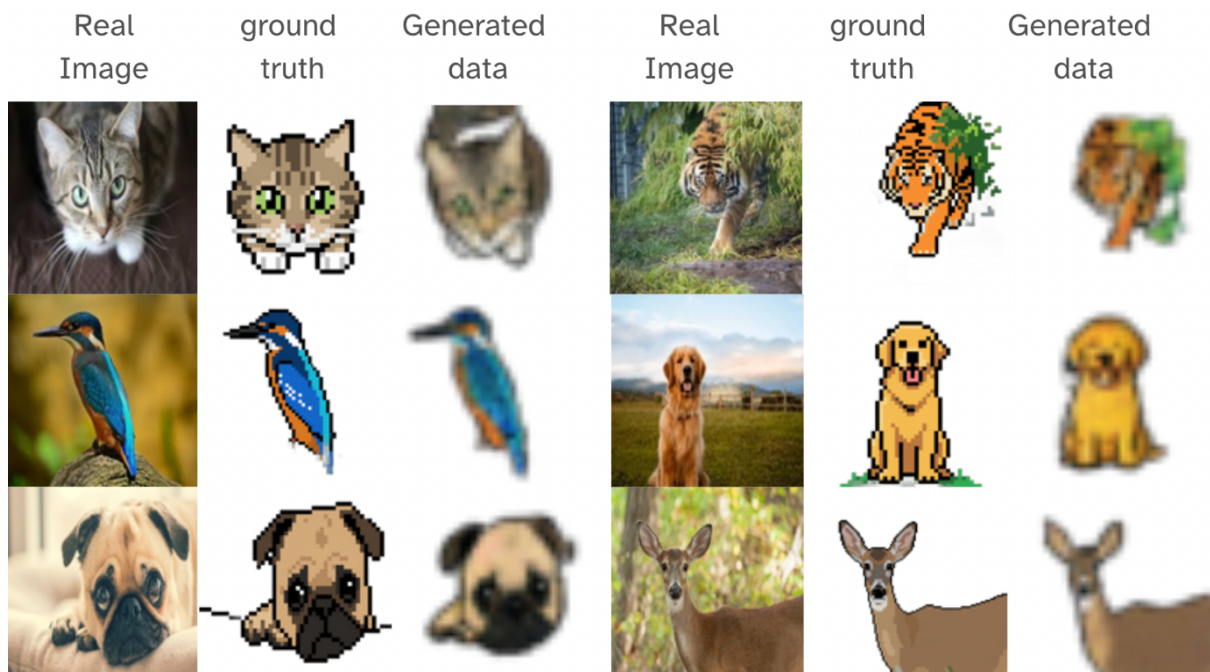


Figure 7: Output of our model

Additionally, Figure 7 presents some examples of the results of our model. By modifying the generation process to upsampling to 32x32, the outputs demonstrate substantial improvements compared to the original pix2pix model. In comparison, our model produces superior outputs which show much clearer boundaries, less blob-like artifacts, and better preservation of object

structure and details. The baseline CNN model produced the poorest outputs, characterized by severe blurring and unrecognizable blob-like structures. The original pix2pix model performed better but still exhibited fragmented boundaries, melted edges, and color collapse. In contrast, our model generated significantly clearer, more complete and structurally integrated pixel-art style images with well-defined boundaries and better details, demonstrating clear superiority over both previous approaches.

4.1.4 Qualitative Evaluation Result

Out of the 100 comparisons:

- 16 candidates preferred the outputs from our modified U-Net model
- 3 candidates preferred the outputs from the baseline CNN model
- 1 candidate preferred the outputs from the original Pix2pix model

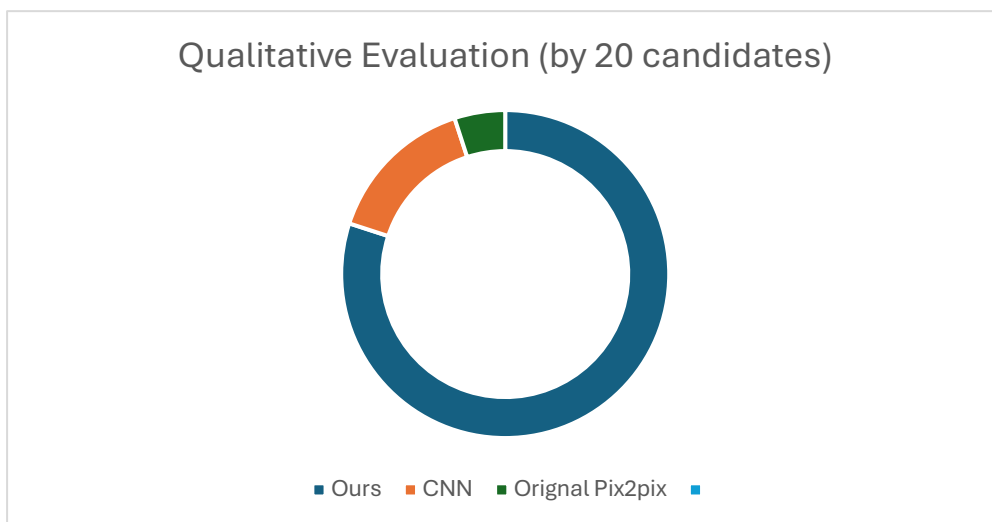


Figure 8: Qualitative Evaluation by 20 candidates (Human Preference Study)

This human evaluation demonstrates a clear preference for our proposed model, indicating that our model — including skip connections, LeakyReLU activation, separate RGB/Alpha output layers, and binary regularization — significantly improve the perceptual quality of the generated pixel art compared to the baseline approaches.

4.2 Quantitative Evaluation Result

4.2.1 Fréchet Inception Distance (FID Score)

FID Score result:

Model	FID Score
Our model	243.5 ± 2.66
Original Pix2pix	212.3 ± 2.18
Baseline CNN	250.4 ± 28.70

FID Score metrics are reported as mean $\pm$ standard deviation over 10 independent training runs with different random seeds.

4.2.2 Learned Perceptual Image Patch Similarity (LPIPS)

LPIPS result:

Model	LPIPS
Our model	0.0484 ± 0.00091
Original Pix2pix	0.0564 ± 0.021
Baseline CNN	0.0524 ± 0.0054

LPIPS metrics are reported as mean $\pm$ standard deviation over 10 independent training runs with different random seeds. The remarkably small standard deviation (0.00091) of our model indicates excellent training stability and robustness, which is particularly important for practical deployment in pixel art generation.

Chapter 5 – Discussion

5.1 Observations Discussion

5.1.1 Why Does Our Model Achieve Better Performance?

The proposed model demonstrates superior performance compared to the original pix2pix model and the baseline CNN model. It generates more ideal results, which are complete and distinctly recognizable structures. In addition, our model reduces fragmentation and blob-like artifacts and still ensures a more stable output structure than the original model. The performance gains are principally due to the architectural strengths of U-Net and the adversarial training mechanism of GAN. The problem which suffering blurry, fragmented or blob-like result on generated images of CNN model is because the downsampling process. About the architecture of the baseline CNN model, due to the compression during repeated downsampling, there are significant loss of image information and details. However, U-Net architecture solves this problem with skip connections. By concatenating feature maps from corresponding encoder layers to the decoder layer, U-Net maintains and recovers the details in the image that would be lost while downsampling. Thus, skip connection enables U-Net to generate the output images with less fragmentation, blob-like artifacts and a more complete structure. Continuing with the architectural aspects of the CNN model and our model, the activation function is different. CNN model commonly employs ReLU which causes dying ReLU problem. Dying ReLU causes inactive units and reduced model capacity.

Piecewise Definition of ReLU:

$$f(x) = \begin{cases} x & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$$

As shown in the definition of ReLU, all negative inputs output zero which is called dying ReLU problem.

Piecewise Definition of LeakyReLU:

$$f(x) = \begin{cases} x & \text{if } x \geq 0 \\ \alpha x & \text{if } x < 0 \end{cases}$$

On the other hand, our model employs Leaky ReLU, where α is a small positive constant. By allowing a small, non-zero gradient for negative inputs, LeakyReLU solves the dying ReLU problem and ensures that all neurons remain trainable to keep image details.

5.1.2 Limitation of FID in Pixel-Style Image Generation

The Fréchet Inception Distance (FID) score result on our model is not ideal which is reasonable and estimable considering the properties of our task. FID evaluates the similarity in latent feature space between two images. Yet, FID is comparing the mean difference and covariance of latent features. Performance will be robust when real and generated images are in the same style. Still, our project focuses on translating real images to pixel-art style images, which is changing the style of the images by reduced resolution. As shown in the result of FID evaluation, the style shift and reduction of resolution cause large differences on both mean and covariance. Hence, the original model is giving out a better FID performance than our model because of producing more realistic images. Therefore, FID may not be the most appropriate approach for evaluating style transfer image generation tasks.

5.1.3 Why is LPIPS a More Suitable Metric for Our Task?

On the other hand, Learned Perceptual Image Patch Similarity (LPIPS) metric performs more ideal and rational results on our task. LPIPS measures perceptual distance through deep features from pre-trained deep neural networks. Those pre-trained deep neural networks have learned a large volume of edges, textures, shapes and semantics from millions of images. Instead of operating raw RGB values directly, LPIPS evaluates similarity on deep features of space. Thus, even though the style shifts and the reduction of resolution in the generated images, LPIPS still consider the ground truth and the generated image as similar, which is like human perception. As evidenced in the results of our evaluation, LPIPS performs a decent result on our model and significantly more appropriate result compared to other models. Therefore, on account of LPIPS align better with human visual perception as well as captures difference on textures, structures and overall appearances that human able to notices, employing LPIPS as the evaluation metric is more suitable than relying on FID.

5.1.4 Suitability of the PatchGAN discriminator

Employing a PatchGAN discriminator, which aggressively penalizes for fine-grained realism, may be not suitable to our intended objective. Since pixel art requires simplified details and restricted colour palettes, a PatchGAN operating on a 32x32 resolution tends to over-focus on edge smoothness and local noise. This often results in the neglect of global consistency and the preservation of the pixel grid. Consequently, for pixel-art style translation task, a discriminator with a larger receptive field or a less intense focus on high-frequency components might be more effective in capturing the essential structural characteristics of the art style.

5.2 Reflection

5.2.1 The Multi-View Challenge

In our initial proposal, generating consistent 8-directional pixel sprites was a core objective. However, as the project progressed, we made the strategic decision to remove the angle-rotating functionality to focus entirely on solving the transparent background and pixel-stylization problems. This scope reduction significantly decreased the engineering difficulty, but the process of making this decision provided profound insights into the realities of deep learning system design.

Initially, we hypothesized that multi-view generation could be achieved simply by concatenating an 8-dimensional one-hot vector (representing N, NE, E, etc.) to the generator's bottleneck layer. Through literature review and early prototyping attempts, we realized this assumption was fundamentally flawed. Adding an angle vector does not magically grant a 2D convolutional network the ability to infer 3D geometry. For example, a network cannot guess the back of a cat if it has only ever been trained on front-facing images, regardless of how the angle is conditioned.

5.2.2 Limitations and Scope of Multi-View Generation

Our findings indicate that true multi-view consistency demands either (1) training on 3D-rendered datasets (e.g., ETH-80) to provide full angular coverage, or (2) complex cross-view contrastive losses to enforce latent-space clustering across viewpoints. We acknowledge that conditional GANs, while exceptional at style translation, are fundamentally limited in geometric extrapolation without explicit 3D supervision. By scoping multi-view generation out of the current study, we focused computational resources on perfecting the single-view RGBA pipeline, ensuring robust pixel-art generation before pursuing multi-view extensions.

5.2.3 Key Insights: Bridging Theory and Practice

Beyond architectural considerations, this project yielded several practical insights rarely documented in academic literature but critical for production-ready stylized image generation.

First, rather than being a simple optimization, loss of balancing in conditional GANs turned out to be a delicate ecosystem. Although the pix2pix objective seems straightforward, excessive L1 weight led the generator to ignore the adversarial signal and produce blurry but structurally safe outputs, while excessive adversarial weight promoted high-frequency pixel noise hallucinations to trick the PatchGAN discriminator and produce chaotic colours.

Second, the ultimate performance limitation turned out to be data quality. Our model internalized the particular artifacts, palette biases, and edge-rendering peculiarities of the web tool used to create ground-truth pairs rather than learning a general mapping from photos to pixel art. This confirms that the model's performance ceiling in paired translation tasks is strictly limited by the reliability and caliber of its training data.

Third, a fundamental disconnect between creative purpose and quantitative measurements was revealed by our comparison of FID and LPIPS. Pixel art is intrinsically "imperfect" by photorealistic standards, with jagged edges, constrained colors, and purposeful simplification. Standard measurements such as FID are intended for photorealistic production. Researchers may optimize the wrong goal if they just use automated metrics. Human-in-the-loop evaluation

is still crucial for stylized generating jobs in creative sectors, as demonstrated by the qualitative human preference study, where our model was favored in 80% of comparisons.

5.3 Future Work

5.3.1 Implement Learned Perceptual Loss for Improving Pixel-Art Sharpness

Although our model successfully generates pixel-style RGBA outputs, the results still have blurry edges. Here is a modification that should reduce these issues. Our current generator loss consists of adversarial loss, L1 reconstruction loss, and binary regularization for alpha layer. While L1 penalizes pixel-wise deviation, it does not capture high-level perceptual similarity. By employing learned perceptual loss based on LPIPS into the training objective, as we discussed LPIPS is trained on human perceptual judgments and has been shown has been demonstrated to exhibit a stronger correlation with visual quality in our image-to-image translation task. Our goal is to maintain higher quality on semantic content and reduce blurry in the generated outputs.

5.3.2 Multi-view Generation

Generating consistent 8-directional pixel sprites from a single real image was identified as Problem 3 in our original proposal. We will set our proposed model be Model A and set Model B to implement multi-view feature. Building upon Model A, developing Model B for 8-directional multi-view generation (N, NE, E, SE, S, SW, W, NW). We will adopt a view-conditioned input encoding approach by injecting an 8-dimensional one-hot vector into the generator. To train Model B, we will utilize dataset, for example ETH-80 or CIFAR-100, which provides multi-view images. Corresponding pixel-art ground truths will be generated using Model A. A cross-view consistency loss will be introduced to ensure style and colour coherence across different views. This two-stage pipeline is expected to enable automatic creation of game-ready 8-directional pixel sprites from a single realistic image, opening new possibilities for game asset generation and stylized content creation.

Chapter 6 – Conclusion

Employing a modified pix2pix conditional GAN framework, this project successfully created an automated pixel art generator with transparent background removal. The two main issues noted in the game industry are immediately addressed by our generator: (1) the absence of specialized tools for producing high-quality pixel-art style generation, and (2) the incapacity of current generators to provide clear RGBA outputs with transparent background appropriate for gaming engines.

Based on the pix2pix architecture, we created a U-Net generator that enforces the clean, grid-constrained appearance necessary for pixel art by upsampling to 32x32 resolution. We implemented two different output layers to manage transparency and RGB channels separately: alpha channel with sharpened sigmoid function scaled by a factor of five and RGB channels with tanh function. Furthermore, the generator loss was modified by implementing a custom binary regularization equation, which effectively forces alpha values toward 0 or 1, which is

fully transparent or opaque, as well as removes color bleed and gray semi-transparent artifacts.

For comparison experiments, a baseline CNN model, which deploys generators with no discriminator and skip connections, was put into place. Our model was clearly superior in both qualitative and quantitative evaluation. The qualitative evaluation result indicates that candidates consider the majority of images to have better results from our model instead of the baseline CNN model and original model. In addition, the quantitative evaluation results indicate that our model achieved significantly better results across different metrics. By achieving a lower LPIPS, demonstrating the effectiveness of implementing pre-trained Deep Neural Network simulates human perception.

As a result, our model produces pre-established pixel art RGBA from realistic images directly, eliminating manual background removal and palette enforcement. This substantially reduces asset production time and cost for indie developers and rapid prototyping in the game industry.

In conclusion, this project demonstrates that production-ready pixel art with transparent backgrounds can be produced by using a properly adjusted pix2pix framework. It highlights the potential of the pix2pix model in optimizing game production processes and offers a useful, effective benchmark for 2D asset creation.

Work Division

Data Collection	Ho Chun Long, Au Pak Yu, Yuen Ching Hin, Chan King Yin, Chan Hoi Lun
Data Preprocessing	Ho Chun Long, Au Pak Yu, Yuen Ching Hin, Chan King Yin, Chan Hoi Lun
Model Building	Ho Chun Long, Chan Hoi Lun
Model Training	Ho Chun Long, Chan Hoi Lun, Chan King Yin
Evaluation	Ho Chun Long, Chan Hoi Lun, Chan King Yin

Reference

- Aman_Kumar094. (n.d.). Animal dataset [Data set]. Kaggle. <https://www.kaggle.com/datasets/amankumar094/animal-dataset>
- Arjovsky, M., Chintala, S., & Bottou, L. (2017). Wasserstein generative adversarial networks. *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 214–223. PMLR. <https://proceedings.mlr.press/v70/arjovsky17a.html>
- Bain & Company. (2025, August). Squeezed in the Middle: AAA gaming studios must adapt – Gaming Report 2025. Retrieved from <https://www.bain.com/insights/squeezed-in-the-middle-aaa-gaming-studios-must-adapt-gaming-report-2025>
- Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y. (2014). Generative adversarial nets. *Advances in Neural Information Processing Systems*, 27, 2672–2680.
- Gulrajani, I., Ahmed, F., Arjovsky, M., Dumoulin, V., & Courville, A. (2017). Improved training of Wasserstein GANs. *Advances in Neural Information Processing Systems (NeurIPS)*, 30. <https://papers.neurips.cc/paper/7159-improved-training-of-wasserstein-gans.pdf>
- Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., & Hochreiter, S. (2017). GANs trained by a two time-scale update rule converge to a local Nash equilibrium. *Advances in Neural Information Processing Systems (NeurIPS)*, 30. <https://arxiv.org/abs/1706.08500>
- Heckbert, P. (1982). Color image quantization for frame buffer display. *ACM SIGGRAPH Computer Graphics*, 16(3), 297–307. <https://doi.org/10.1145/800248.807402>
- Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33, 6840–6851. <https://proceedings.neurips.cc/paper/2020/file/4c5bcfec8584af0d967f1ab10179ca4b-Paper.pdf>
- Isidore, C. (2016, March 18). Sega’s video-game artists had to draw pixels by hand on crazy custom machines in the 1980s and 90s. Quartz. <https://qz.com/766684/segas-video-game-artists-had-to-draw-pixels-by-hand-on-crazy-custom-machines-in-the-1980s-and-90s>
- Isola, P., Zhu, J.-Y., Zhou, T., & Efros, A. A. (2017). Image-to-image translation with conditional adversarial networks. In *Proceedings of 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 5967–5976). IEEE. <https://doi.org/10.1109/CVPR.2017.632>
- Kingma, D. P., & Welling, M. (2014). Auto-encoding variational Bayes. *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/1312.6114>

- Maas, Andrew L., Awni Y. Hannun, and Andrew Y. Ng. "Rectifier nonlinearities improve neural network acoustic models." *Proc. icml*. Vol. 30. No. 1. 2013.
- Mao, X., Li, Q., Xie, H., Lau, R. Y. K., Wang, Z., & Smolley, S. P. (2017). Least squares generative adversarial networks. *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2813–2821.
https://openaccess.thecvf.com/content_ICCV_2017/papers/Mao_Least_Squares_Generative_ICCV_2017_paper.pdf
- McKinsey & Company. (2024, April). What is deep learning? Retrieved from <https://www.mckinsey.com/featured-insights/mckinsey-explainers/what-is-deep-learning>
- Mildenhall, B., Srinivasan, P. P., Tancik, M., Barron, J. T., Ramamoorthi, R., & Ng, R. (2020). NeRF: Representing scenes as neural radiance fields for view synthesis. *European Conference on Computer Vision (ECCV)*. <https://arxiv.org/abs/2003.08934>
- Mirza, M., & Osindero, S. (2014). Conditional generative adversarial nets. *arXiv preprint from* <https://arxiv.org/pdf/1411.1784>
- Miyato, T., Kataoka, T., Koyama, M., & Yoshida, Y. (2018). Spectral normalization for generative adversarial networks. *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/1802.05957>
- Nair, V., & Hinton, G. E. (2010). Rectified linear units improve restricted Boltzmann machines. In *Proceedings of the 27th International Conference on Machine Learning (ICML)*, 807–814. <https://www.cs.toronto.edu/~fritz/absps/reluICML.pdf>
- Nichol, A. Q., & Dhariwal, P. (2021). Improved denoising diffusion probabilistic models. *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 8162–8171. PMLR. <https://proceedings.mlr.press/v139/nichol21a/nichol21a.pdf>
- Shinkle, K. (2021, July 5). *How indie games are expanding beyond pixel art*. Screen Rant. <https://screenrant.com/indie-games-pixel-art-2d-hand-drawn-3d/>
- Wang, Z., Bovik, A. C., Sheikh, H. R., & Simoncelli, E. P. (2004). Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4), 600–612. <https://doi.org/10.1109/TIP.2003.819861>
- Xu, N., Price, B., Cohen, S., & Huang, T. S. (2017). Deep image matting. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. <https://arxiv.org/abs/1703.03872>
- Yoo, I., Luo, X., Wang, Y., Yang, F., & Milanfar, P. (2020). GIFnets: Differentiable GIF encoding framework. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. https://openaccess.thecvf.com/content_CVPR_2020/papers/Yoo_GIFnets_Differentiable_GIF_Encoding_Framework_CVPR_2020_paper.pdf
- Zhang, R., Isola, P., Efros, A. A., Shechtman, E., & Wang, O. (2018). The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE*

Conference on Computer Vision and Pattern Recognition (CVPR).

<https://arxiv.org/abs/1801.03924>

Zhu, J.-Y., Park, T., Isola, P., & Efros, A. A. (2017). Unpaired image-to-image translation using cycle-consistent adversarial networks. *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2242–2251. IEEE.

<https://doi.org/10.1109/ICCV.2017.244>

Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. (2022). High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 10684–10695).

<https://arxiv.org/abs/2112.10752>